

Pavel Vasilyev · markdown

Software engineer & verification architect — LLM oversight, grounding, and the infrastructure to scale them

OPEN TO WORK

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PROFILE

I work as the human in the loop on LLM and agent systems: reviewing what models produce, grounding outputs in verified sources, and making claims accountable before they ship. I build the verification harnesses and oversight flows that let this scale. Fifteen years across applied statistics, ML infrastructure, and developer tooling — on GitHub since 2010, with 150+ projects. The hard part is not the typing; it is deciding precisely what must exist. **Spec before code.**

WHAT I DO

- **LLM & agent oversight** — independently verify outputs (not spot-check); run adversarial passes to refute claims; push coverage until new findings stop. Outcome: teams ship high-confidence LLM features.
- **Grounding & provenance** — trace every claim to its source; make every artifact auditable (repo, commit, agent, session). Outcome: stakeholders know exactly what they're trusting.
- **Applied statistics & machine learning** — time-series and statistical modelling, ML pipelines, and the discipline to know when a result is real versus an artefact of method. Outcome: decisions backed by defensible data.
- **Spec-first tool-building** — build verification harnesses, inference gateways, and oversight scaffolds. Outcome: teams iterate on reasoning, not tuning config.

EXPERIENCE — selected highlights, not a complete history

LLM Engineer — Move38 2026 (contract)

Company: US-based early-stage startup making tactile electronic games and educational technology.

- Shipped an LLM feature end-to-end: collaborated with designers on player-facing interactions, architected serving infrastructure, tuned prompts for engagement, ensured low-latency experience. Single-handedly took it from infrastructure to live product.

Dissemblage.art — self-directed art practice 2024 — 2025

Practice: Exploring language models as creative material — testing their outputs, discovering failure modes, and shaping raw generations into coherent artistic work.

- Ongoing LLM-aided experiments probing what models actually produce versus their marketing; documenting edge cases and demonstrating human oversight as creative material, not a band-aid for model failure.

Chief Technology Officer — LevEhad 2024

Organization: *Israeli NGO providing services and advocacy around disability inclusion.*

- Owned IT infrastructure and technology strategy end-to-end; partnered with mission and operations to reduce friction for staff and serve accessibility requirements for clientele.

AI Architect — Stamina AI

2023

Company: *Mid-sized distributed team building AI-powered customer support automation.*

- Architected LLM systems that reduced customer-facing hallucinations: designed ontology-backed grounding, implemented semantic retrieval, and iterated with product/support teams to fit real workflows and expectations.

Research Lead — Spring Research

2020 — 2022

Company: *Quantitative research firm focused on financial market modeling and predictive infrastructure.*

- Led research team detecting market regimes and identifying trading windows; designed and deployed ML pipelines (AWS/SageMaker) on multi-modal time series for low-latency inference; partnered with infrastructure and trading teams on production deployments.

Independent practice — consulting & open-source development

2010 — present

Scope: *Consulting, advisory, and software practice between salaried roles.*

- Built tools for clients and teams on specs they provided; continuous open-source work (150+ projects, eight+ languages, all spec-first). Domains: verification workflows, LLM harnesses, audio processing, type theory. See *Selected work* below.

Earlier roles — data science, NLP & software engineering

2009 — 2019

Scope: *Data scientist, NLP engineer, and software roles across Tel Aviv, Hong Kong, Prague — Nestlogic, Maverick Medical AI, Athena Portfolio Solutions, Ten-X, Verba, Finfo.*

- Financial-news NLP and signal extraction; medical-text classification and information extraction; computer vision pipelines; corpus-linguistics engines and text-analysis infrastructure.

SELECTED WORK — open source, github.com/marchdown

- co-loop — *scaffold for agentic collaboration: fork an agent's work, branch it, loop changes back; no manual merge conflicts*
- spec-before-code — *Rust · encode your spec as invariants before coding; catch misalignment early*
- topo-sonify — *Rust · sonify topographic data; explore terrain through sound*
- nixbloom — *Nix · ephemeral, reproducible development workspaces on NixOS*
- Full GitHub profile — 150+ projects across eight+ languages

SKILLS & TOOLS

Languages: Swift · Rust · Clojure · Elixir / Erlang · Haskell · Nix · TypeScript · Agda · C (GDScript / C++ when the platform insists)

Domains: applied statistics & machine learning · LLM & agent orchestration · oversight & verification · developer tooling · audio & sonification · type theory & proof assistants

EDUCATION

MSc Computational Linguistics — RSUH, Moscow 2015

BSc Nuclear Physics — Czech Technical University, Prague 2009

Ongoing: Self-directed study in type theory, proof assistants, and language design. Published work: [agda-proofs](#), [harmony](#) — pushing the formal foundations of verification and program correctness.

LANGUAGES

English · Hebrew · Russian — working knowledge of German & French.

CONTACT & COLLABORATION

Email	uprootiny@gmail.com
GitHub	github.com/marchdown
Web	hyperstidious.org
Collaborate	co-loop — fork the seed, run your agent, we loop

marchdown.github.io · github.com/marchdown/marchdown.github.io

CV assembled 2026-07-17 · built from verifiable GitHub data + owner-confirmed roles/degrees